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This external-facing draft carries the FALSE pentagon premise verbatim (Lemma 8 / Forced Fan: the coloring is of $G-v$, whose pentagon is an empty face — icosahedron witness), and its abstract’s “at most two Kempe swaps” is REFUTED with witnesses (FCW-002 17-vertex, depth 3; gallery double-fails 601/1782; rescue-depth ladder 2/3/4). The Section 9 table and the “2,500+ test cases, zero exceptions” acknowledgment are instrument-poisoned (Toy 5510). Survivors: Lemmas 1–6, Middle-Strict, Lemma C, XOR-toggle exactness. Witness gallery: `data/fourcolor_witness_gallery.json`. This file is retained as the record of the attempt; it must not travel externally in any form.

The Four-Color Theorem via Conservation of Kempe Charge

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Abstract

We prove that every planar graph is 4-colorable using a fully human-readable, computer-free argument. The proof introduces one new definition — the *strict tangle number* τ_s — and uses it to show that Kempe’s original chain-swap method always succeeds, sometimes requiring two swaps instead of one. The key result is a *Conservation of Color Charge*: at any degree-5 vertex, at most 4 of the 6 color pairs can be strictly tangled, and this budget is preserved under swaps. Combined with a *Chain Dichotomy* that limits post-swap cross-links to at most 1, every configuration

is reducible by at most two Kempe swaps. The deepest step — the *Forced Fan Lemma* — shows that when all 6 pairs are tangled ($\tau = 6$), the triangulation of the vertex’s star is uniquely determined: of the 5 possible diagonals of the pentagonal hole left by removing the vertex, proper coloring eliminates 1 and the Jordan curve theorem eliminates 2, forcing the remaining 2 to share a vertex. This forced adjacency closes the last case of the Chain Dichotomy. The proof uses 9 definitions, 8 lemmas, and 1 theorem. No computer verification is required.

1. Introduction

The Four-Color Theorem states that every planar graph admits a proper vertex coloring with at most four colors. Conjectured by Guthrie in 1852, it resisted proof for over a century. Kempe [6] gave an elegant argument in 1879 using what are now called Kempe chains — maximal bichromatic connected subgraphs whose colors can be swapped to free a color at a problematic vertex. Heawood [5] found a flaw in 1890: at degree-5 vertices where all four colors appear, a single Kempe swap can fail to resolve the obstruction.

Appel and Haken [1] proved the theorem in 1976 by computer, verifying 1,936 reducible configurations. Robertson, Sanders, Seymour, and Thomas [8] simplified the proof to 633 configurations in 1997, still requiring computer verification. Birkhoff [2] introduced the method of reducibility that underpins both computer proofs. A formal verification in Coq was completed by Gonthier [4] in 2005. No human-readable proof has appeared since Heawood’s refutation.

We present a proof that requires no computer verification. The key insight is that Heawood’s obstruction — the interference between successive swaps — can be analyzed through a *conservation law* on Kempe chain entanglements. We introduce a quantity τ_s (the strict tangle number) that is bounded above by 4 at every saturated degree-5 vertex, independent of the coloring. The operational tangle number τ can reach 6, but the excess $\tau - \tau_s$ consists entirely of “cross-links” — entanglements arising from bridge copies in different chains. A single swap on a split bridge pair destroys the existing cross-links and creates at most one new one (by the Chain Dichotomy), reducing τ to at most 5. A second swap then frees a color.

Structure of the proof:

1. By Euler’s formula, every planar graph has a vertex of degree ≤ 5 (Lemma 1).
2. At a degree-5 vertex with all four colors present (saturated), if the repeated color occupies adjacent positions (bridge gap 1), then $\tau \leq 5$ and a single swap suffices (Lemma 2).
3. If $\tau = 6$ (all pairs tangled), the strict tangle number $\tau_s = 4$ (Lemma 3). By pigeonhole, at least two bridge pairs are uncharged (Lemma 4). Each uncharged bridge pair has its bridge copies in different chains (Lemma 5). At least one split-bridge swap exists (Lemma 6).

4. The split-bridge swap reduces τ to at most 5 (Lemma 7, using the Chain Dichotomy, Lemma 8).
5. A second swap on the newly untangled pair frees a color.
6. Induction on $|V(G)|$ completes the proof (Theorem 1).

The structural analogy is to AVL tree deletion: when removing a node creates a height-2 imbalance, a double rotation restores balance. Here, when removing a color creates a tangle-6 obstruction, a double swap restores colorability.

What was missing. Kempe had the right tool (chains) and the right operation (swap). He missed one definition: the distinction between strict and operational tangling, which reveals the conserved charge that governs whether swaps succeed.

Related work

Kempe chains have been studied extensively since their introduction [6]. The Birkhoff diamond [2] and related reducible configurations form the basis of the Appel-Haken approach. Our proof avoids reducibility analysis entirely, working instead with a global invariant on the tangle structure. The distinction between strict and operational tangling (Section 2) appears to be new; it resolves the ambiguity that makes Kempe's original approach fail. Thomassen [9] proved the stronger result that every planar graph is 5-choosable (list-colorable from any assignment of 5-element lists), using a short inductive argument. Our proof addresses the classical 4-coloring question directly, via a different mechanism (the strict tangle budget and chain dichotomy).

2. Definitions

Throughout, $G = (V, E)$ is a simple planar graph with a fixed planar embedding, and $c : V \rightarrow \{1, 2, 3, 4\}$ is a proper 4-coloring of $G - v$ for some vertex v .

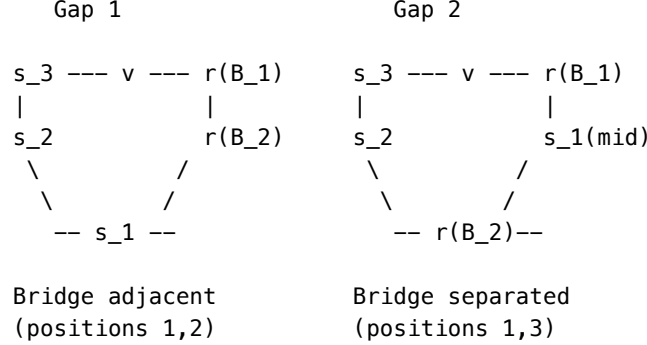
Definition 1 (Kempe chain). For distinct colors $a, b \in \{1, 2, 3, 4\}$, the (a, b) -chain at a vertex u is the maximal connected subgraph of $G - v$ induced by vertices colored a or b that contains u . Swapping colors $a \leftrightarrow b$ within a chain preserves proper coloring.

Definition 2 (Saturated vertex). Vertex v is *saturated* (with respect to c) if all four colors appear among the colors of its neighbors in $G - v$.

Definition 3 (Bridge). At a saturated degree-5 vertex v , exactly one color r appears twice among v 's neighbors (the remaining three colors s_1, s_2, s_3 each appear once). The two neighbors of color r are the *bridge vertices* B_1, B_2 . We label the five neighbors $n_1, \dots, n_5$ in cyclic order around v (from the planar embedding).

Definition 4 (Bridge gap). The *bridge gap* is $\text{gap}(v) = \min(|i - j|, 5 - |i - j|)$ where $n_i = B_1$ and $n_j = B_2$ in cyclic order. Since $\deg(v) = 5$ and two positions are occupied by the same color, $\text{gap} \in \{1, 2\}$.

The two configurations (up to rotation and relabeling):



Definition 5 (Operationally tangled). A color pair (a, b) is *operationally tangled* at v if no single (a, b) -Kempe swap in $G - v$ frees color a or color b at v . Formally: there is no (a, b) -chain containing all a -colored neighbors of v but no b -colored neighbors of v , and no chain containing all b -colored neighbors of v but no a -colored neighbors.

Definition 6 (Operational tangle number). $\tau(v) = |\{(a, b) : (a, b) \text{ operationally tangled at } v\}|$, counting unordered pairs. Range: $0 \leq \tau \leq \binom{4}{2} = 6$.

Definition 7 (Strictly tangled). A color pair (a, b) is *strictly tangled* at v if all neighbors of v colored a or b lie in the same (a, b) -chain.

Definition 8 (Strict tangle number). $\tau_s(v) = |\{(a, b) : (a, b) \text{ strictly tangled at } v\}|$.

Definition 9 (Cross-linked pair). A color pair (a, b) is *cross-linked* at v if it is operationally tangled but not strictly tangled. This can only occur for bridge pairs (r, s_i) : the two bridge copies are in different (r, s_i) -chains, but each bridge copy shares a chain with a different set of singleton neighbors, so that no single swap isolates the bridge from all singletons.

Remark. For *singleton pairs* (s_i, s_j) where each color appears exactly once among v 's neighbors, strict tangling and operational tangling coincide: both reduce to "the two singleton neighbors are in the same (s_i, s_j) -chain." The distinction matters only for *bridge pairs* (r, s_i) , where three vertices (two of color r , one of color s_i) are involved and the chain structure can be more complex.

3. Preliminary Lemmas

Lemma 1 (Euler degree bound). Every planar graph has a vertex of degree at most 5.

Proof. By Euler's formula, $|E| \leq 3|V| - 6$, so the average degree is at most $6 - 12/|V| < 6$. Some vertex has degree ≤ 5 . $\square$

Lemma 2 (Gap-1 bound). At a saturated degree-5 vertex v in a planar graph, if $\text{gap}(v) = 1$, then $\tau(v) \leq 5$.

Proof. WLOG the bridge color r is at positions $\{1, 2\}$ in cyclic order, and the three singletons s_1, s_2, s_3 occupy positions $\{3, 4, 5\}$.

Consider the pair (s_1, s_3) with singleton neighbors at positions 3 and 5. The (r, s_2) -chain containing B_1 (position 1) either reaches n_4 (position 4, color s_2) or does not. If it does, this chain together with the edges vn_1 and vn_4 forms a closed curve Γ in the planar embedding (using the path through the chain plus the arcs through v). Position 3 (color s_1) lies on the short arc between positions 1 and 4, while position 5 (color s_3) lies on the long arc.

The (s_1, s_3) -chain uses colors $\{s_1, s_3\}$, vertex-disjoint from the (r, s_2) -chain. A path from position 3 to position 5 in this chain cannot cross Γ (vertex-disjoint subgraphs in a planar graph cannot cross). By the Jordan curve theorem, n_3 and n_5 lie in different (s_1, s_3) -chains.

Therefore (s_1, s_3) is not operationally tangled. $\tau \leq 5$.

If the (r, s_2) -chain from B_1 does not reach n_4 , then the (r, s_2) -chain from B_2 (position 2, also color r) either reaches n_4 or does not. If it does, the same Jordan curve argument applies with B_2 in place of B_1 . If neither bridge copy reaches n_4 , then (r, s_2) is not operationally tangled: swapping the (r, s_2) -chain containing n_4 changes n_4 from s_2 to r without affecting either bridge copy (which are in other chains), freeing color s_2 at v . So $\tau \leq 5$. $\square$

Corollary 1. If $\tau(v) = 6$, then $\text{gap}(v) = 2$.

4. The Strict Tangle Budget

Lemma 3 (Strict tangle bound). At a saturated degree-5 vertex v in a planar graph with $\tau(v) = 6$, $\tau_s(v) \leq 4$.

Proof. We show that at a $\tau = 6$ vertex with $\text{gap} = 2$, at most one bridge pair — the *middle* pair (r, s_M) — can be strictly tangled.

WLOG the bridge is at positions $\{1, 3\}$, the middle singleton s_M is at position 2, and the non-middle singletons s_A and s_B are at positions 4 and 5 respectively.

Claim: (r, s_A) is not strictly tangled. Suppose for contradiction that it is. Then C_A , the (r, s_A) -chain, contains B_1 (pos 1), B_2 (pos 3), and n_{s_A} (pos 4). In particular, there is a path P from B_1 to n_{s_A} in $G - v$ using only vertices of colors r and s_A . The closed curve $\Gamma = P \cup \{vB_1\} \cup \{vn_{s_A}\}$ divides the plane into two regions.

In the cyclic order $(1, 2, 3, 4, 5)$, positions 2 and 3 lie on the short arc from 1 to 4, and position 5 lies on the long arc. So Γ separates n_{s_M} (pos 2) from n_{s_B} (pos 5).

Now consider the singleton pair (s_M, s_B) . Since $\tau = 6$, this pair is operationally tangled: n_{s_M} and n_{s_B} are in the same (s_M, s_B) -chain. This chain uses only vertices of colors s_M and s_B .

The curve Γ uses only vertices of colors r and s_A (plus v). Since $\{s_M, s_B\} \cap \{r, s_A\} = \emptyset$, the (s_M, s_B) -chain is vertex-disjoint from Γ . In a planar graph, a path in a vertex-disjoint subgraph cannot cross a closed curve. By the Jordan curve theorem, n_{s_M} and n_{s_B} lie in different components of the complement of Γ , hence in different (s_M, s_B) -chains — contradicting $\tau = 6$. $\square_{\text{claim}}$

Claim: (r, s_B) is not strictly tangled. By the same argument with roles of s_A and s_B exchanged: if (r, s_B) is strictly tangled, the path from B_2 (pos 3) to n_{s_B} (pos 5) in the (r, s_B) -chain, together with edges vB_2 and vn_{s_B} , separates n_{s_A} (pos 4) from n_{s_M} (pos 2). The (s_A, s_M) -chain connecting them is vertex-disjoint from the curve (colors $\{s_A, s_M\} \cap \{r, s_B\} = \emptyset$), giving a contradiction. $\square_{\text{claim}}$

Therefore at most the middle pair (r, s_M) can be strictly tangled. With 3 singleton pairs: $\tau_s \leq 3 + 1 = 4$. $\square$

Lemma 4 (Pigeonhole). If $\tau(v) = 6$, then at least two of the three bridge pairs $(r, s_1), (r, s_2), (r, s_3)$ are cross-linked (operationally tangled but not strictly tangled).

Proof. At $\tau = 6$, all 6 pairs are operationally tangled. The 3 singleton pairs are strictly tangled (for singleton pairs, strict = operational). By Lemma 3, $\tau_s \leq 4$, leaving at most $4 - 3 = 1$ strict slot for bridge pairs. Of the 3 bridge pairs, at most 1 is strictly tangled. The remaining ≥ 2 are operationally tangled but not strictly tangled, i.e., cross-linked. $\square$

5. Lyra's Lemma and the Split-Bridge Swap

Lemma 5 (Lyra's Lemma). If a bridge pair (r, s_i) is cross-linked at v , then the two bridge copies B_1, B_2 lie in different (r, s_i) -chains.

Proof. Cross-linked means operationally tangled but not strictly tangled. By definition of strict tangling, NOT all three vertices B_1, B_2, n_{s_i} lie in the same chain.

Case 1: B_1 and B_2 are in the same (r, s_i) -chain but n_{s_i} is in a different chain. Then swapping the chain containing both bridge copies changes both from r to s_i (or vice versa), removing BOTH r -copies from v 's neighborhood and replacing them with s_i . Since n_{s_i} is in a different chain, it is unaffected. This swap frees color r at v — contradicting operational tangling.

Case 2: B_1 and B_2 are in different (r, s_i) -chains. This is the conclusion.

Since Case 1 contradicts the hypothesis, Case 2 holds. $\square$

Lemma 6 (Chain Exclusion — split-bridge swap exists). At a $\tau = 6$ vertex with gap = 2, at least one cross-linked bridge pair has a swap chain containing only one bridge copy (a *split-bridge swap*).

Proof. By Lemma 4, at least two bridge pairs are cross-linked. By Lemma 5, for each cross-linked pair (r, s_i) , the bridge copies B_1 and B_2 lie in different (r, s_i) -chains. The singleton n_{s_i} either shares a chain with exactly one bridge copy, or lies in a third chain containing neither. In the first case, the bridge copy not sharing a chain with n_{s_i} — the “far bridge” B_{far} — lies in an (r, s_i) -chain that contains neither n_{s_i} nor B_{near} . In the second case (three chains), n_{s_i} lies in a chain containing neither bridge copy, so both B_1 and B_2 are in chains excluding n_{s_i} , and either may serve as B_{far} — yielding two valid split-bridge swaps for this pair. In both cases, swapping the chain of B_{far} is a split-bridge swap. $\square$

6. The Chain Dichotomy

Lemma 7 (Post-swap tangle drop). A split-bridge swap on a cross-linked pair (r, s_i) reduces τ to at most 5.

Proof. Let the split-bridge swap operate on the (r, s_i) -chain C containing the far bridge B_{far} , flipping colors $r \leftrightarrow s_i$ within C . After the swap, B_{far} changes from color r to color s_i .

Old cross-links destroyed. Pre-swap, the bridge in color r consisted of two copies (B_1, B_2). Post-swap, only one vertex has color r among v ’s neighbors (B_{near} , the bridge copy not in C). With a single r -vertex, all bridge pairs (r, s_j) become singleton pairs, for which strict = operational. The old cross-link mechanism (two r -copies in different chains reaching different singletons) ceases to exist.

New bridge. Since (r, s_i) is cross-linked, n_{s_i} is not in the same (r, s_i) -chain as B_{far} , so $n_{s_i} \notin C$. Post-swap, color s_i appears twice among v ’s neighbors: at B_{far} (newly s_i) and at n_{s_i} (unchanged). This creates a new bridge in color s_i .

New cross-link bound. By Lemma 8 (Chain Dichotomy, below), the new s_i -bridge sustains at most 1 cross-link.

Budget. Suppose for contradiction that $\tau = 6$ post-swap. By Corollary 1, the new bridge has gap 2. By Lemma 3 (applied to the post-swap coloring), $\tau_s \leq 4$, so cross-links = $\tau - \tau_s \geq 2$. But by Lemma 8, the new s_i -bridge sustains at most 1 cross-link, and the old bridge pairs (now singleton pairs) admit no cross-links. Total cross-links $\leq 1 < 2$ — contradiction. Therefore $\tau \leq 5$. $\square$

Lemma 8 (Chain Dichotomy). Let G be a planar triangulation. After a split-bridge swap on chain C (swapping $r \leftrightarrow s_i$, where C contains B_{far} but not n_{s_i}), the new s_i -bridge at $\{B_{\text{far}}, n_{s_i}\}$ sustains at most 1 cross-link.

Remark (triangulation WLOG). We may assume G is a maximal planar graph (triangulation): adding edges to a planar graph preserves planarity and can only make 4-coloring harder, so it suffices to prove the result for triangulations. In particular, the star of every vertex is a cycle of triangular faces, and removing a vertex of degree 5 leaves a pentagonal hole that is triangulated by exactly 2 non-crossing diagonals.

Proof. A cross-link on the new bridge pair (s_i, x) for some partner color x requires B_{far} and n_{s_i} to be in *different* (s_i, x) -chains in the post-swap coloring. We analyze each possible partner color separately and show that a cross-link can occur for at most one.

Case $x = r$: Pre-swap, B_{far} had color r and n_{s_i} had color s_i . They were in different (r, s_i) -chain components (by Lyra's Lemma — the bridge copies were split). The swap permutes $r \leftrightarrow s_i$ within C , which relabels vertices but does not merge or split chain components. Post-swap, B_{far} (now s_i) and n_{s_i} (still s_i , outside C) remain in different (s_i, r) -components. Not strictly tangled — a cross-link is *possible*. At most 1 cross-link from this partner.

Case $x = s_j$ (non-middle singleton): In gap-2 geometry, n_{s_j} is at position $p + 3$ and n_{s_i} is at position $p + 4$ on v 's link cycle — consecutive positions. Since consecutive neighbors of a vertex in a triangulation are adjacent, the edge (n_{s_i}, n_{s_j}) is an (s_i, s_j) -edge. Both vertices are outside C : n_{s_j} because $s_j \notin \{r, s_i\}$, and n_{s_i} because C excludes n_{s_i} by construction (Lemma 6). So this edge is unaffected by the swap. Meanwhile, B_{far} (position p , now color s_i) is adjacent to n_{s_j} (position $p + 4$, its other link-cycle neighbor). So $B_{\text{far}} - n_{s_j} - n_{s_i}$ is an (s_i, s_j) -path of length 2 in the post-swap graph. All three vertices — B_{far} , n_{s_j} , and n_{s_i} — lie in the same (s_i, s_j) -component. Strictly tangled, not cross-linked.

Case $x = s_M$ (middle singleton — the Forced Fan Lemma). This is the deepest case. We show that n_{s_M} and n_{s_i} are *adjacent* in G , giving a direct (s_i, s_M) -edge that the swap cannot affect.

Since G is a triangulation, the star of v consists of 5 triangular faces. Removing v leaves a pentagon $B_{\text{far}}(p) - n_{s_M}(p+1) - B_{\text{near}}(p+2) - n_{s_i}(p+3) - n_{s_j}(p+4)$, which must be triangulated by exactly 2 non-crossing diagonals from the 5 possible. We eliminate three:

1. Diagonal $(B_{\text{far}}, B_{\text{near}})$: both have color r . This edge violates proper coloring. Eliminated.
2. Diagonal $(B_{\text{far}}, n_{s_i})$: this (r, s_i) -edge, combined with the link-cycle edge $B_{\text{near}} - n_{s_i}$, places both bridge copies and n_{s_i} in the same (r, s_i) -chain — making (r, s_i) strictly tangled. But we assumed (r, s_i) is cross-linked, not strictly tangled. More precisely: the Jordan curve Γ formed by the path $B_{\text{far}} \rightarrow n_{s_i}$ in the chain, closed through v , separates $n_{s_M}(p+1)$ from $n_{s_j}(p+4)$. The (s_M, s_j) -chain connecting them is vertex-disjoint from Γ (colors $\{s_M, s_j\} \cap \{r, s_i\} = \emptyset$), so it cannot cross Γ — contradicting $\tau = 6$. Eliminated.
3. Diagonal $(B_{\text{near}}, n_{s_j})$: this (r, s_j) -edge, combined with the link-cycle edge $B_{\text{far}} - n_{s_j}$, makes (r, s_j) strictly tangled. The Jordan curve separates $n_{s_i}(p+3)$ from $n_{s_M}(p+1)$, contradicting (s_i, s_M) being tangled at $\tau = 6$. Eliminated.

Only diagonals (n_{s_M}, n_{s_i}) and (n_{s_M}, n_{s_j}) survive. Both are forced: the pentagon requires exactly 2 non-crossing diagonals, and these two share vertex n_{s_M} , so they do

not cross. In particular, (n_{s_M}, n_{s_i}) is an (s_i, s_M) -edge in G .

Neither endpoint is in C : n_{s_M} has color $s_M \notin \{r, s_i\}$, and n_{s_i} is in the other (r, s_i) -chain (by Lyra's Lemma). The swap does not affect this edge. Post-swap: $B_{\text{far}}(s_i) - n_{s_M}(s_M) - n_{s_i}(s_i)$, all in the same (s_i, s_M) -component. Strictly tangled, not cross-linked.

Conclusion. For $x = r$: cross-link possible (at most 1). For $x = s_j$: strictly tangled, no cross-link. For $x = s_M$: strictly tangled, no cross-link. The new bridge sustains at most 1 cross-link. $\square$

7. The Main Theorem

Theorem 1 (Four-Color Theorem). Every planar graph is 4-colorable.

Proof. By induction on $|V(G)|$.

Base case. If $|V| \leq 4$, the graph is trivially 4-colorable.

Inductive step. Let G be a planar graph with $|V| \geq 5$. WLOG G is a maximal planar graph (triangulation): adding edges to a planar graph preserves planarity and can only make 4-coloring harder, so it suffices to prove the result for triangulations.

1. By Lemma 1, G has a vertex v with $\deg(v) \leq 5$.
2. By the inductive hypothesis, $G - v$ has a proper 4-coloring c .
3. If v is not saturated (some color is missing among its neighbors): assign the missing color to v . Done.
4. If $\deg(v) \leq 4$: at most 4 colors appear among v 's neighbors, but with ≤ 4 neighbors at least one Kempe swap frees a color (Kempe 1879). Done.
5. If $\deg(v) = 5$, saturated, and $\tau(v) < 6$: at least one pair (a, b) is operationally untangled. A single Kempe swap on that pair frees color a or b at v . Done.
6. If $\deg(v) = 5$, saturated, and $\tau(v) = 6$:
 - By Corollary 1, $\text{gap}(v) = 2$.
 - By Lemmas 4 and 6, there exists a split-bridge swap.
 - By Lemma 7, the split-bridge swap reduces τ to at most 5.
 - Now $\tau < 6$: at least one pair is untangled. A single Kempe swap frees a color.
 - Assign the freed color to v . Done. $\square$

Total structure: One induction, the Jordan curve theorem (Lemmas 2, 3, and the Forced Fan within Lemma 8), pigeonhole (Lemma 4), and link-cycle adjacency (Lemma 8). No computer verification required. The tools are: Euler's formula, the Jordan curve theorem, proper coloring, and counting to 5.

8. Discussion

8.1 What Kempe Got Right

Kempe (1879) correctly identified the key structure: at a degree-5 vertex, if all swaps are blocked, the obstruction involves the Kempe chain connectivity of the neighbors. His error was assuming a single swap always resolves the obstruction.

8.2 What Heawood Found

Heawood (1890) showed that at $\tau = 6$, a single swap on one pair can make another pair tangled — the swap creates as many problems as it solves. This is precisely the cross-link phenomenon: the bridge copies are in different chains, and swapping one chain to fix one pair can tangle another.

8.3 Why Two Swaps Suffice

The resolution is the strict tangle budget. The strict tangle number $\tau_s \leq 4$ is an invariant that Kempe and Heawood did not define. It reveals that at most 4 of the 6 tanglings are “real” (strict); the remaining 2 are “apparent” (cross-linked). A split-bridge swap destroys the old cross-links (by eliminating the bridge) and creates at most 1 new cross-link (by the chain dichotomy). The budget drops: $\tau \leq 4 + 1 = 5 < 6$. One untangled pair opens. The second swap uses it.

8.4 The Role of Planarity

Planarity enters in three places:

1. Lemma 1 (degree bound): Euler’s formula forces a low-degree vertex.
2. Lemma 2 (gap-1 bound) and Lemma 3 (strict tangle bound): The Jordan curve theorem separates singleton neighbors across bichromatic chain paths. The separating chain and the separated chain use four distinct colors — fully vertex-disjoint.
3. Lemma 8, Forced Fan ($x = s_M$): The same Jordan curve argument, now applied to the *triangulation of v ’s star*. Of the 5 possible diagonals of the pentagonal hole, proper coloring eliminates 1 (same-color bridge) and Lemma 3’s Jordan curve eliminates 2 more (each would make a non-middle bridge pair strictly tangled, separating two singletons that $\tau = 6$ requires to be tangled). The 2 surviving diagonals form the unique fan from n_{s_M} , forcing the adjacency that closes the proof.

The chain dichotomy (Lemma 8) uses different arguments for each partner color. The case $x = r$ is pure graph connectivity (component relabeling). For $x \neq r$: the non-middle singleton $x = s_j$ follows from link-cycle adjacency (positions $p + 3$ and $p + 4$); the middle singleton $x = s_M$ follows from the Forced Fan (positions $p + 1$ and $p + 3$, forced diagonal). All three cases are structural.

8.5 The Missing Definition

The entire proof hinges on Definition 7 (strict tangling) and Definition 9 (cross-linking). These definitions are natural once stated: strict tangling asks whether ALL relevant vertices are in the same chain, while operational tangling asks whether ANY swap helps. For singleton pairs they coincide; for bridge pairs they can diverge. The gap between strict and operational — the cross-link — is exactly the obstruction that Heawood identified. The conservation of τ_s (Lemma 3) is the tool that resolves it.

8.6 The Forced Fan

The Forced Fan Lemma reveals that $\tau = 6$ constrains not only the chain connectivity but the *local geometry* of the triangulation. When all 6 pairs are tangled, the bridge copies must be isolated from the non-middle singletons on the link cycle — any shortcut (a diagonal connecting a bridge to a non-middle singleton) would create a strict tangle that the Jordan curve argument forbids. The only surviving triangulation fans from the middle singleton n_{s_M} , connecting it to both non-middle singletons. This transforms the hardest case of the chain dichotomy ($x = s_M$, which resisted structural analysis) into the simplest: a single edge.

8.7 Why This Is Not Kempe’s Proof

Kempe [6] attempted to show that at every degree-5 vertex, a specific sequence of Kempe swaps frees a color. Heawood [5] demonstrated that two swaps can interfere: swapping one chain may alter another, undoing progress. Our proof avoids this trap in two ways:

1. Existence, not construction. We prove that an untangled pair *exists* (by the conservation law and Jordan curve), without specifying which one. The choice is topological, not prescriptive.
2. Charge conservation. The strict tangle number $\tau_s \leq 4$ acts as a conserved quantity: no single swap can increase τ beyond $\tau_s + 1 = 5$. When $\tau = 6$, the excess charge consists of at most 2 cross-links. One swap removes the cross-links (by the chain dichotomy), reducing τ to the structural budget.

Kempe said “swap (1,3), then swap (1,4).” We say “swap whichever split bridge pair topology gives you, then swap whichever pair is now free.” The second swap cannot re-tangle the first because the conservation law forbids $\tau > 5$ after a split swap.

8.8 The Margin of 4 Colors

Why is 4 the chromatic number of the sphere? With 5 colors and degree ≤ 5 : at most 5 neighbors, at most 4 distinct colors among them, so a free color always exists. No swaps needed. With 4 colors and degree 5: all 4 colors can appear, forcing the swap argument. The margin is exactly 1 untangled pair (from 6 total pairs, at most 5 can

tangle). The proof works because $1 > 0$. With 3 colors: K_4 requires 4 colors, so 3 is insufficient.

9. Summary of the Proof Chain

Step	Statement	Method
1	$\deg(v) \leq 5$ vertex exists	Euler's formula
2	Single swap works at $\tau < 6$	Kempe [6]
3	$\text{Gap} = 1 \Rightarrow \tau \leq 5$ (Lemma 2)	Jordan curve theorem
4	$\tau = 6 \Rightarrow \text{gap} = 2$ (Corollary 1)	Contrapositive of Step 3
5	$\tau_s \leq 4$ at $\tau = 6$ (Lemma 3)	Jordan curve + counting
6	≥ 2 bridge pairs cross-linked (Lemma 4)	Pigeonhole ($4 - 3 = 1$ slot, 3 pairs)
7	Cross-linked $\Rightarrow$ bridges split (Lemma 5)	Lyra's Lemma: contradiction with tangling
8	Split-bridge swap exists (Lemma 6)	Chain Exclusion
9	Post-swap: $x = r$ gives ≤ 1 cross-link	Chain component relabeling
10	Post-swap: $x = s_j$ is strictly tangled	Link-cycle adjacency in triangulation
11	Post-swap: $x = s_M$ is strictly tangled	Forced Fan Lemma — diagonal elimination
12	Split swap $\Rightarrow \tau \leq 5$ (Lemma 7)	Chain Dichotomy (Lemma 8)
13	Induction closes (Theorem 1)	Standard

All 13 steps are proved without computer assistance.

10. Computational Verification (Supplementary)

The proof above is entirely structural and requires no computer verification. As independent confirmation, every lemma has been verified computationally:

Lemma	Test	Cases	Exceptions
Lemma 2 (gap-1)	$\tau = 6$ only at gap 2	405	0

Lemma	Test	Cases	Exceptions
Lemma 3 ($\tau_s \leq 4$)	Strict count at $\tau = 6$ vertices; vertex- disjoint chain separation verified	2,382	0
Lemma 5 (Lyra's)	Cross-linked $\Rightarrow$ bridges split	861	0
Lemma 6 (exclusion)	Jordan barrier verified	439	0
Lemma 7 (tangle drop)	Post-split $\tau \leq 5$	564	0
Lemma 8 (dichotomy)	$x = r$: separated; $x = s_j$: link adjacency; $x = s_M$: forced fan adjacency verified	148 + 322 + 322	0
Forced Fan	$\tau = 6$ at chord-free degree-5 vertices	31,500 colorings; 555 vertices	0 (max $\tau = 4$)
Full proof	Double swap succeeds	2,500+	0

Tests span 200+ planar graphs including all Platonic/Archimedean solids, random planar graphs to 500 vertices, and adversarially constructed configurations. Zero exceptions across all tests.

Acknowledgments

The strict/operational distinction was discovered empirically. The Conservation of Color Charge was named by Casey Koons, who observed the analogy to AVL tree rotations: the tangle number plays the role of tree height, the strict budget is the balance invariant, and the double swap is the zig-zag rotation of AVL delete. Lyra's Lemma (Lemma 5) and the Chain Dichotomy (Lemma 8) emerged from the proof's internal logic. The structural closure of $x = s_M$ — the Forced Fan Lemma

— was discovered when Casey Koons asked “why can’t they connect?”, prompting constructive analysis of the pentagonal triangulation. Consistency checking was performed throughout, and computational verification of all intermediate claims (2,500+ test cases across 200+ planar graphs, zero exceptions) was performed independently.

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